

Designing Augmented Reality Based Interventions to Encourage Physical Activity During Virtual Classes

Manjul Singh Sachan

School of Information, Rochester Institute of Technology
Rochester, NY, USA
ms5687@rit.edu

Roshan L. Peiris

School of Information, Rochester Institute of Technology
Rochester, NY, USA
roshan.peiris@rit.edu

ABSTRACT

This paper presents an exploratory study of designing augmented reality (AR) based interventions to encourage physical activity among students attending virtual classes. We conducted a focus group study with four HCI students to understand the current behaviour of students during breaks between classes. Based on these insights we designed two AR interventions: AR Exergame, an interactive game that requires the user to move their hands to grab virtual apples; AR micro-movements, a method that requires the user to switch to a different physical space when starting a new virtual class. These AR interventions were aimed at encouraging students to perform micro-activities in between virtual classes, i.e. small non-strenuous activities. The effectiveness of these methods to reduce video conferencing fatigue, a.k.a Zoom fatigue, and also to compare it with the students current methods was tested through a user study with six participants.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality; Interaction devices.**

KEYWORDS

AR, Zoom fatigue, interventions

ACM Reference Format:

Manjul Singh Sachan and Roshan L. Peiris. 2022. Designing Augmented Reality Based Interventions to Encourage Physical Activity During Virtual Classes. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3491101.3519749>

1 INTRODUCTION AND RELATED WORK

Since the beginning of the COVID-19 pandemic, the use of video-conferencing platforms has increased drastically and they have become a necessary tool for conducting virtual classes and meetings remotely. Although these platforms have helped us to stay

connected with our friends, family and colleagues, and have enabled educational institutions to continue classroom learning virtually, the effects of their increased usage and engagement are not all positive [10]. There are an increasing number of studies that have identified the negative effects of virtual classes and meetings known as “Zoom Fatigue” - the exhaustion felt due to prolonged usage of video-conferencing platforms (VCP) which exerts physical and mental toll on the users [1, 9, 26]. Moreover, attending virtual classes from home requires the students to engage with VCPs over extended periods of time which leads to an increase in Sedentary Behavior and a decrease in Physical Activity (PA) [25]. Numerous studies have shown that high levels of sedentary behavior are associated with diseases like type II diabetes, cardiovascular diseases, and breast cancer, which in-turn can have negative psychological effects and can become a cause of depression [6, 19].

Additionally, the virtual classroom environment has a limited support to emulate a physical classroom and there are fewer possibilities for physical activity[4]. For example, while attending school/college in-person, students have interim period of time between classes to physically move around in their environment - to switch classrooms or to participate in social activities, while the same behavior is not exhibited or sometimes not possible while attending virtual classes. One way to overcome this is through promoting Exergames (Exercise Games) as part of the students' routine which have shown substantial results in decreasing sedentary behavior among inactive children [16, 19]. However, physical activity and sedentary behavior are not mutually exclusive as the potential benefits of physical activity could be undone if the students spend the remaining hours of their day sitting in front of their computers [11]. Hence, there is a need to promote *micro* health interventions by integrating them into everyday tasks without disrupting ongoing routine [20]. These are health interventions which blend into the periphery of users' attention and can enable them to follow regular physical activity exercises.

Bakker et al. suggest that for peripheral interactions to quickly and frequently shift between the users' center and periphery of attention, the interaction needs to be easy to initiate [2]. This is in accordance with the findings of Mueller et al. [18] and Mandryk et al. [17] who present that the design of casual exergames should also provide an “easy entry into play”. Combining these two research areas, previous research has explored the possibility of Augmented Reality (AR) exergames that enable the users to move physically while gamifying their experience to provide incentives [24]. However, most of these systems require the use of gaming consoles like Xbox Kinect or Nintendo Wii, which are expensive and thus not feasible for every household environment [17]. In addition, Shin et al. implemented a webcam based design using AI to recognise hand

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519749>

and body movements to interact with games like "Pop the Balloons" and "Name Body Parts" which were used as a part of a separate virtual physical education class [23]. Inspired by these works, we explore using AR to provide micro health interventions seamlessly (as activities in between virtual classes) and how effective they are to mitigate Zoom fatigue among students.

In this research, we aim to investigate the existing methods that college students use for overcoming Zoom fatigue and explore the design and effectiveness of an AR intervention system. We conducted a preliminary semi-structured focus group with HCI students to inform the design and methodology of a subsequent user study where we simulate a virtual classroom environment with micro-activities using an AR system. Here we define micro-activities as short duration low-intensity movements (e.g. similar to moving from class to class in an in-person environment) which encourage physical activity among students, similar to micro-break activities suggested by Kim et al. to recover from daily work demands [15]. The user study involved three tasks where the participants watched short lecture videos and performed micro-activities in between two videos. A detailed description of the study is provided in the methodology section below. The system is evaluated through a mixed-methods study design where we collect and analyze quantitative data about the level of exhaustion the participants feel throughout the three tasks along with qualitative data about their overall experience of using the systems and their feedback for improving the preliminary design. Our results showcase the possibility of using AR to reduce sedentary behavior among college students during virtual classes and we further discuss the design implications and considerations in future prototypes of the system.

2 METHODOLOGY AND RESULTS

We conducted this research study in two parts. In the first part, we invited HCI students to take part in a semi-structured focus group study where we initially discussed our problem statement in brief and then guided an open-ended discussion between the participants. The ideas and insights gained from the focus group discussion became the basis for next part of the research and it informed the procedure design of the user study. In this section, we present the research process and results for both the studies in detail.

2.1 Focus Group Study

2.1.1 Participants and Research Setup. The participants for the focus group study were recruited through advertisements and word of mouth on campus. All of the participants were students who were pursuing Master's in HCI with prior experience in user research and study design process. In total, 4 participants (3 female 1 male, mean age 25) were present and the discussion was conducted virtually over Zoom video-conferencing platform which lasted for approximately 1 hr 15 min. The participants were requested to provide consent and were given \$10 as compensation.

2.1.2 Procedure. The discussion started with the researcher giving a brief overview about the purpose of the study, followed by a semi-structured interview process to keep the conversation focused and open-ended. The main purpose of the discussion was to gain insights about the participants' current behaviours to mitigate

physical stress and exhaustion while engaged in prolonged virtual classes or meetings and thereafter brainstorm ideas to design intervention systems in the micro-activity design space. Half-way through the discussion, the participants were asked to take a short 5-min break which acted as a stimulus for them to discuss and reflect on their own behavior and the type of activities they did during the break. This discussion was recorded and then transcribed. We codified the quotes from the transcripts and performed a bottom-up (inductive) thematic analysis [3, 14] on these quotes.

2.1.3 Results and Discussions. Based on the results of thematic analysis on the participant responses and through prior understanding and literature review of the subject matter by the researchers, we identified three overarching themes in the behaviour of students during their breaks in between virtual classes – small non-strenuous activities (walking, stretching etc.); changing environments (e.g. moving to a different location); switch focus (e.g. browse social media during breaks)

Small non-strenuous activities: We found that participants often preferred small scale activities (e.g. micro activities) that are not physically exhausting. Two participants would usually *"stand up and stretch"* or *"take a small walk"*. However, one participant also mentioned that *"doing mundane tasks like drinking water or using the bathroom doesn't feel relaxing"*.

Changing environments: When asked about how the environment they take their virtual class in would affect their experience, they said that *"it is a huge factor"*. It even affects how they feel when taking a break, for example *"moving to a different room to take a break feels significantly different than staying in the same room to take a break"* and that *"changing location might not be as beneficial than doing an activity"*.

Switching focus: All of the participants also like to switch focus away from the classroom during short breaks and they all agree that *"switching focus might be better than completely detaching from work"* and that *"detachment from the system would be required for it to feel like a break"*. These were indicated in the participant comments where 1 participant would *"keep sitting at [their] desk and play games on [their] smartphone"* and another would sometimes *"listen to a song"*. These findings showcase a need for a system which would help the students to shift their focus to a different task while still being present in the classroom environment.

When brainstorming ideas for implementing a micro-health intervention system, the participants propose that *"it can suggest you things to do in 5 minutes based on your environment"* and that the type of activities will *"depend on what resources you have around you to take a break"*. Additionally, they want the system to provide *"easy work mode to break mode switching"* while having the *"ability to delay or snooze the break alert"* with *"user defined level of control"*.

Based on these recommendations from the participants, we came up with the idea for a system that can directly integrate into the students' workstations and work in the periphery to provide interventions for taking breaks. The system can suggest micro-activities to do during the breaks, like walking or changing locations, based on the students' environment and preferences. The need for contextual awareness of the student's environment can be realized through Augmented Reality [13] which not only enables us to detect the context but also can introduce interactive elements for them to engage

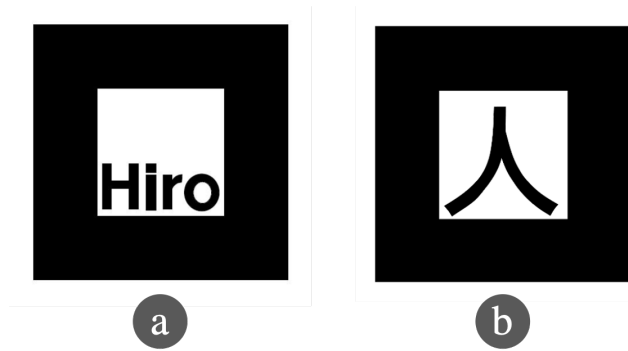


Figure 1: Markers used for the AR Exergame and the AR Micro-activities applications: (a) A Hiro AR Marker used to load 3D model of an apple for the exergame. (b) A Kanji AR Marker used as an additional marker to define a new space for AR micro-activities.

with during their breaks. These interactive elements will act as the intervention to encourage physical activity and would change based on the students' environment. Additionally, as a part of our goal we wanted the system to have a low-barrier to entry/easy to adopt and designed it to reduce the common hindrances in initiating physical activities like temporal, physical and psychological barriers [17]. Therefore, using AR based interventions could possibly integrate into the user's environment and work schedules, thus reducing the former two barriers, while the gamification aspect would help to overcome the latter.

2.1.4 AR Interventions. For the purpose of our research, we developed two interventions which utilise AR to suggest micro-activities to the students. First one is an AR exergame which promotes physical activity through moving and stretching the body, while the second one is about encouraging the students to switch their location or environment in-between virtual classes or during breaks. Below we present prototypes for implementing these interventions and test their effectiveness through a subsequent user study.

We built these interventions using the AR.js JavaScript library in which we detect AR markers (Figure 1) in the user's environment and display the corresponding interactions associated with the marker. For our preliminary investigations, we served the game over a web browser which can be accessed using any device that has a built-in camera. This setup enabled us to use a laptop with an integrated webcam to conduct the entire study and did not require the use of AR gaming consoles like Kinect or Wii. In addition, we expect that these methods can be integrated with existing VCPs such as Zoom (and using the device integrated cameras) so that routine interventions can be introduced seamlessly in user's schedule.

AR Exergame: We modelled this intervention based on the design of a popular game called "GrabApple" where the player has to catch the falling apples on the screen [12]. When the user scans the Hiro marker (Figure 1(a)), the system starts to display a 3D model of an apple at random points on the screen, changing the position at 2 second intervals. To play the game, players have to move their

hands towards the position of the apple before it disappears from the screen, and they are encouraged to move and stretch their body to reach the extreme positions. The game was implemented as a Wizard-of-Oz prototype in which the users were given full freedom to interact with the system in any way they want and the gamification was enabled through manually recording the score for the number of apples they grabbed within the task duration. To make the game more engaging when playing over a period of time, we introduced a gamified feature where the camera feed from the webcam was laterally flipped (mirror image) to make the players' moves a bit obscure.

AR Micro-movements: The goal of this method was to explore encouraging the user to do a quick movement in between virtual classes or virtual meetings (similar to moving from one classroom to another or one meeting room to another in in-person settings). As such, when the user attempts to join the next virtual meeting or class, the user is required to have a different marker scanned or appear in their background. By design, the user will have to keep the markers in different physical locations of their space. For example, the user may have the markers set up on two different walls or on opposite ends of the same wall that requires a small physical movement (Figure 2(b)). In this intervention, when the user scans the second marker such as the Kanji marker (Figure 1(b)), the system displays an alert on the web browser which contains a URL that can link to any lecture, meeting or video, and the link would be unique to this particular marker only. By updating the the new meeting links that are binded with the AR markers, we aimed to emulate switching of the classroom environments for the students.

2.2 User Study

In this study we aimed to evaluate the effectiveness of the AR interventions against Zoom fatigue through a user study [1]. The ultimate goal of the study was to simulate a virtual classroom environment and introduce different micro-activities (from hereon referred as activities) in between lectures (or videos) to understand how these activities affect the students' sedentary behaviour and also to see how the context of their environment affects their attention and focus during classes.

2.2.1 Study Design. We designed a within-subject evaluation for conducting the user study which consisted of the following three activities as the main independent variable:

- (1) AR Exergame: Playing the AR exergame to encourage students to do some PA, like moving and stretching.
- (2) AR micro-movement: Switching to a different location and scan AR marker to access the next video.
- (3) Short break (baseline): Taking a short break to do any activity that the students usually perform during breaks in their virtual class routines.

The dependent variable was the perceived overall fatigue that was measured with the standard Zoom Exhaustion & Fatigue Scale (ZEF) (only using the General and Visual Fatigue questionnaires) [8] and the third activity served as the baseline treatment against which we compare the effects of the AR intervention activities. The three activities were randomized for each participant using the Latin Square method [22] and each activity was performed twice



Figure 2: AR Interventions for reducing zoom fatigue (a) AR Exergame: The system detects the AR marker in the background and uses it as a reference to display ‘virtual apples’ (shown inside the dotted circle) in random locations. The user is required reach out and grab the apples as they appear (b) **AR Micro-movements:** (b-1) The user has several AR markers in the physical working space (encircled by white circles). To enter a virtual class, a new/different marker is required to be visible in the background. To enter the next virtual class, the user is required to move to a different location such as the other side of the table so that a different unique marker is visible. (b-2) The user has physically moved to different location to enter the next virtual class.

within each task. Each task consisted of watching two 10-minute instructional videos and performing one activity after each video. Therefore, each participant completed 3 tasks where they watched 6 videos and performed 6 activities in total. After each task the participants were asked to fill out the ZEF questionnaire to collect quantitative data about their perceived exhaustion level. The responses were recorded on a 5-point Likert scale with values ranging from “Not at all” to “Extremely”. Finally, at the end of the study, each participant was interviewed to gather qualitative data about their overall experience and feedback.

2.2.2 Participants and Research Setup. We recruited 6 participants (3 female, 3 male, mean age 25) for the study through advertisements on campus. The study was conducted in a lab setting with a desk and chair to simulate virtual classroom environment and an AR marker was pasted on the wall behind the chair which would be visible on the webcam. The same laptop device was used for every participant which had the necessary software (Node.js and Chrome Web Browser) installed to run the application.

2.2.3 Procedure. After a brief introduction to the purpose and design of the study, the participants were asked to provide their consent for participating in the study and give permission for collecting data through audio recording. Each participant was assigned a number which would be recorded with the data and no personally identifiable information was stored. The experimenter then did the preliminary setup which required calibration of the laptop screen to the AR marker position. This setup was required so that the 3D models appear within the screen limits and do not overflow out of the display. After the setup, participant was shown the first 10-minute instructional video for the first task, after which they were asked to perform the first randomly chosen activity. The duration for “AR Exergame” and “Short Break” activities was kept at 5-minutes, while the duration for “AR Micro-movements” activity was defined by the time it took for the participants to reach and scan the second AR Marker. After the activity, the participant watched a second 10-minute video and performed the same activity again, followed by filling out the ZEF questionnaire for the respective task. This process was repeated two more times for Task 2 and 3

and afterwards the experimenter conducted a post-study interview with the participant. In the post study interview, each participant was asked about their overall experience of doing the 3 activities and their recommendations for features they would like to have in both the AR interventions to improve the engagement and increase their motivation to interact with them in the future. Their answers are discussed in the discussions and future work section below. The full study took approximately 1 hr 30 min for each participant.

2.2.4 Data Analysis. The quantitative data was compiled and analysed to identify the Overall fatigue experienced by each participant for each task. The tasks were assigned a number according to the activity that was performed in each task. Thus, Task 1 was identified by the AR exergame activity, Task 2 was short break activity and Task 3 corresponded to AR micro-movement activity. The Likert scale responses for the ZEF questionnaire were converted into numerical values of 0-4 from “Not at all” - “Extremely” respectively, and Mean and Standard Deviation was calculated for each task. The qualitative data was gathered from transcribing the interview responses and a deductive thematic analysis [3, 14] on the participants’ quotes was performed to identify specific themes relating to the AR interventions and overall study. The themes were reviewed by the second author and we identified 5 overarching themes: Preference on activities, Design recommendation for an AR application, Social gamification and acceptability, Effects of classroom environment, and Overall feedback.

2.2.5 Results and Discussions. As the Figure 3 shows, the Overall Fatigue for all the tasks was close to or lower than 1 which corresponds to the feeling of “Slight” fatigue on the ZEF scale. Even though the overall fatigue was lowest for the baseline task where participants took a short break to perform any activity which they usually do in their routine, qualitative results show that the participants would prefer to interact with the AR interventions in their breaks, especially with the improvement of their features. While two participants did say “I would prefer to take 5 min break and doing some other activity”, one of them mentioned that “depending on how AR/VR evolves, I would like to play that between classes”. For one of the participants, the preference would “depend on [their] mood”

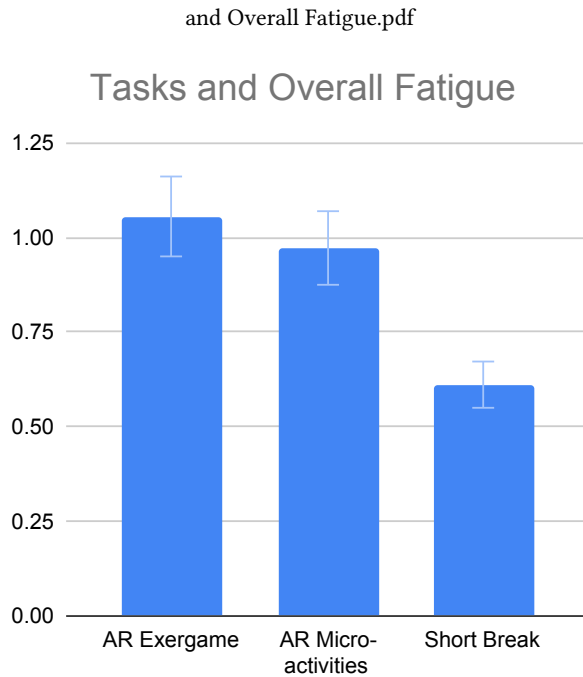


Figure 3: Bar graph showing the level of Overall Fatigue felt by participants after each task.

and the availability of space in the environment. While another participant mentions that they “know playing the AR game would be better for [them] than just taking a break and doing nothing”.

These results indicate the willingness of students to adopt AR based micro-health interventions into their virtual class routines and with the evolution of this technology, more students would be interested to take part. This being a preliminary study conducted over a short period of time, the quantitative data sample was not enough to effectively measure the exhaustion and fatigue levels for the 3 activities. Thus, it would require a much larger scale longitudinal study to achieve contrasting results between the different activities.

The participants were intrigued by the AR exergame and were interested to share feedback on how the game can be improved and be made more engaging. All of the participants said that they would prefer to play AR exergames in the future if it allows them to detach from the virtual class to do a “fun” activity. They gave feedback on the existing design of the game, with one participant mentioning “[they] felt more involved because of the mirrored effect”. However, many participants made design recommendations to improve the “GrabApple” game in an AR application as well, like “adding different interactions like [slashing gesture] in Fruit Ninja game”, “make apples harder to reach to perform better stretches”, and “adding different elements like kicking a ball to engage other body parts”. Many participants would also like to have a social gamification aspect to the game where they can “play group games with friends” or “share scores on social media” and “[Play] something

that helps to engage the whole class as that is lacking in virtual classes”.

This result is in line with the research findings of previous studies by Yu Chen and Pearl Pu! [5] and Ren et al. [21], who show that players like to participate in cooperative fitness games and they feel more motivated in groups. However, one participant mentioned that not all games should be social as they prefer to play games “without scores and competition” and in these cases it would be beneficial to explore designs for puzzle or one-person arcade games. In the future, we can explore different AR games which integrate with the video-conferencing platforms and allows students to either play single-player exergames, like GrabApple [12], or compete against each other in collaborative exergames, like Play Volleyball [23].

Concerns of social acceptability of these AR exergames were also raised as not all students would be comfortable in performing the required tasks in public places or even at home with other roommates around. Two participants voiced these concerns, saying that “If someone sees me doing the activity it would be very weird” and “I will be shy or conscious to do these activities in public space like library”. This shows that more research is needed to test the effects of exergames in public setting and to also explore the ethical aspects of sharing physical activity data among classmates.

We observed that most of the participants would like to move around after long periods of sitting down and mention that it helps them to pay more attention and focus on their lecture, which aligns with the meta-analysis research findings of de Greeff et al. where acute physical activity shows a positive effect on attention [7]. Identifying with AR Micro-movements, one participant mentioned “I prefer to move around in my environment to be active and I focus better”, while another said “I prefer standing for some time after a long session of sitting for lectures”. While doing the task 3, some participants chose to switch to a more comfortable seating to watch the second video, stating that “it helps me to pay better attention”. Some participants also mentioned how the features already implemented in current VCPs help them to switch focus and move around, like “With camera off, I feel comfortable in moving around and switching places” and “Breakout rooms during virtual classes facilitate conversations, but they are again online”. However, as the previous quote mentions, it is difficult to detach from the work entirely as these features are implemented in the system and some times not even utilised. Thus we identify that AR micro-movements method could be further enhanced to encourage students to move around during virtual classes such as when moving to a break-out room or do micro activities in between classes when there is not enough time to take a longer break.

2.3 Conclusion and Future Work

With our design of AR intervention system, we have observed the possibility of serving AR interventions directly on web browser and using the in-built webcam of the device which is available in most commercially available laptop computers used by students to attend virtual classes. As such, we recognize that these methods can be directly integrated with VCPs and integrated directly to schedules, etc. Additionally, we identify that our AR micro-activity system can also be used to demarcate “working”, “active” or “relaxing” zones in a users environment, which can also be improved in the future

using Indoor Localization techniques [27]. Future systems can even cease the reliance on AR markers and use image recognition and classification such as natural feature tracking to detect the context of the user's environment to serve different peripheral micro-health interventions.

Our study was a preliminary exploration and thus it presents several limitations which can be addressed in future research. First, it was conducted with a small set of participants in a lab setting and was focused on gathering initial insights. It lacks the detailed results and analysis that can be gained through a long-term in-situ experiment. Secondly, we do not utilise data collection devices like wearable health trackers or sedentary behavior sensing furniture, to track data about the participants' physical activity and exhaustion levels, and rely primarily on their self-evaluation and qualitative feedback. Health tracking data can be utilised in future studies to quantitatively measure how different micro-activities affect the exhaustion and fatigue levels in students. Finally, in future explorations, we can design and build AR interventions that provide more engaging experiences and also incorporate social gaming features.

REFERENCES

- [1] Jeremy N. Bailenson. 2021. Nonverbal Overload: A Theoretical Argument for the Causes of Zoom Fatigue. 2, 1 (Feb. 2021). <https://doi.org/10.1037/tmb0000030>
- [2] Saskia Bakker, Elise van den Hoven, and Berry Eggen. 2015. Peripheral interaction: characteristics and considerations. *Personal and Ubiquitous Computing* 19, 1 (Jan. 2015), 239–254. <https://doi.org/10.1007/s00779-014-0775-2>
- [3] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [4] Kevin G. Byrnes, Patrick A. Kiely, Colum P. Dunne, Kieran W. McDermott, and John Calvin Coffey. 2021. Communication, collaboration and contagion: “Virtualisation” of anatomy during COVID-19. *Clinical Anatomy* 34, 1 (Jan. 2021), 82–89. <https://doi.org/10.1002/ca.23649>
- [5] Yu Chen and Pearl Pu. 2014. HealthyTogether: exploring social incentives for mobile fitness applications. In *Proceedings of the Second International Symposium of Chinese CHI on - Chinese CHI '14*. ACM Press, Toronto, Ontario, Canada, 25–34. <https://doi.org/10.1145/2592235.2592240>
- [6] Ida Damen, Hans Brombacher, Carine Lallemand, Rens Brankaert, Aarnout Brombacher, Pieter van Wesemael, and Steven Vos. 2020. A Scoping Review of Digital Tools to Reduce Sedentary Behavior or Increase Physical Activity in Knowledge Workers. *International Journal of Environmental Research and Public Health* 17, 2 (Jan. 2020), 499. <https://doi.org/10.3390/ijerph17020499>
- [7] Johannes W. de Greeff, Roel J. Bosker, Jaap Oosterlaan, Chris Visscher, and E. Hartman. 2018. Effects of physical activity on executive functions, attention and academic performance in preadolescent children: a meta-analysis. *Journal of Science and Medicine in Sport* 21, 5 (May 2018), 501–507. <https://doi.org/10.1016/j.jsams.2017.09.595>
- [8] Geraldine Fauville, Mufan Luo, Anna Carolina Muller Queiroz, Jeremy N. Bailenson, and Jeff Hancock. 2021. *Zoom Exhaustion & Fatigue Scale*. SSRN Scholarly Paper ID 3786329. Social Science Research Network, Rochester, NY. <https://papers.ssrn.com/abstract=3786329>
- [9] G. Fauville, M. Luo, A. C. M. Queiroz, J. N. Bailenson, and J. Hancock. 2021. Zoom Exhaustion & Fatigue Scale. *Computers in Human Behavior Reports* 4 (Aug. 2021), 100119. <https://doi.org/10.1016/j.chbr.2021.100119>
- [10] Noritoshi Fukushima, Masaki Machida, Hiroyuki Kikuchi, Shiho Amagasa, Toshio Hayashi, Yuko Odagiri, Tomoko Takamiya, and Shigeru Inoue. 2021. Associations of working from home with occupational physical activity and sedentary behavior under the COVID-19 pandemic. *Journal of Occupational Health* 63, 1 (2021), e12212. <https://doi.org/10.1002/1348-9585.12212>
- [11] Yue Gao, Kathrin M. Gerling, Regan L. Mandryk, and Kevin G. Stanley. 2014. Decreasing sedentary behaviours in pre-adolescents using casual exergames at school. In *Proceedings of the first ACM SIGCHI annual symposium on Computer-human interaction in play (CHI PLAY '14)*. Association for Computing Machinery, New York, NY, USA, 97–106. <https://doi.org/10.1145/2658537.2658693>
- [12] Yue Gao and Regan Mandryk. 2012. The Acute Cognitive Benefits of Casual Exergame Play. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Austin, Texas, USA) (CHI '12). Association for Computing Machinery, New York, NY, USA, 1863–1872. <https://doi.org/10.1145/2207676.2208323>
- [13] Jens Grubert, Tobias Langlotz, Stefanie Zollmann, and Holger Regenbrecht. 2016. Towards pervasive augmented reality: Context-awareness in augmented reality. *IEEE transactions on visualization and computer graphics* 23, 6 (2016), 1706–1724.
- [14] Michelle E Kiger and Lara Varpio. 2020. Thematic analysis of qualitative data: AMEE Guide No. 131. *Medical teacher* 42, 8 (2020), 846–854.
- [15] Sooyeol Kim, YoungAh Park, and Qikun Niu. 2017. Micro-break activities at work to recover from daily work demands. *Journal of Organizational Behavior* 38, 1 (2017), 28–44.
- [16] Regan L. Mandryk and Kathrin M. Gerling. 2015. Discouraging sedentary behaviors using interactive play. *Interactions* 22, 3 (April 2015), 52–55. <https://doi.org/10.1145/2744707>
- [17] Regan L. Mandryk, Kathrin M. Gerling, and Kevin G. Stanley. 2014. Designing Games to Discourage Sedentary Behaviour. In *Playful User Interfaces: Interfaces that Invite Social and Physical Interaction*, Anton Nijholt (Ed.). Springer, Singapore, 253–274. https://doi.org/10.1007/978-981-4560-96-2_12
- [18] Florian Mueller, Rohit Ashok Khot, Kathrin Gerling, and Regan Mandryk. 2016. Exertion Games. *Foundations and Trends® in Human-Computer Interaction* 10, 1 (2016), 1–86. <https://doi.org/10.1561/11000000041>
- [19] Phuong Nguyen, Long Khanh-Dao Le, Dieu Nguyen, Lan Gao, David W. Dunstan, and Marj Moodie. 2020. The effectiveness of sedentary behaviour interventions on sitting time and screen time in children and adults: an umbrella review of systematic reviews. *International Journal of Behavioral Nutrition and Physical Activity* 17, 1 (Sept. 2020), 117. <https://doi.org/10.1186/s12966-020-01009-3>
- [20] Xipei Ren, Pengcheng An, Tilde Bekker, Yu Chen, Rohit Ashok Khot, Martijn ten Bhömer, Yunlong Wang, and Gabriele Spina. 2020. Weaving Healthy Behaviors into New Technology Routines: Designing in (and for) the COVID-19 Work-from-Home Period. In *Companion Publication of the 2020 ACM Designing Interactive Systems Conference*. ACM, Eindhoven Netherlands, 393–396. <https://doi.org/10.1145/3393914.3395911>
- [21] Xipei Ren, Lotte Hollander, Rylana van der Marel, Lieke Molenaar, and Yuan Lu. 2019. Step-by-Step: Exploring a Social Exergame to Encourage Physical Activity and Social Dynamics among Office Workers. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–6. <https://doi.org/10.1145/3290607.3312788>
- [22] John T.E. Richardson. 2018. The use of Latin-square designs in educational and psychological research. <https://www.sciencedirect.com/science/article/pii/S1747938X18301623>
- [23] Soohyun Shin, Jaekyung Cho, and Seong-Woo Kim. 2021. Jumble: Interactive Contents for the Virtual Physical Education Classroom in the Pandemic Era. In *Augmented Humans Conference 2021*. ACM, Rovaniemi Finland, 268–270. <https://doi.org/10.1145/3458709.3458964>
- [24] Misha Sra and Chris Schmandt. 2015. Design Strategies for Playful Technologies to Support Light-intensity Physical Activity in the Workplace. *arXiv:1512.02921 [cs]* (Dec. 2015). <http://arxiv.org/abs/1512.02921> arXiv: 1512.02921.
- [25] Universidad de Los Lagos, Jairo Azócar-Gallardo, Universidad de Castilla-La Mancha, Alex Ojeda-Aravena, Universidad de Los Lagos, and Universidad de Castilla-La Mancha. 2022. Active breaks during virtual classrooms in schoolchildren in times of COVID-19 pandemic: A strategy to increase physical activity levels. *Journal of movement & Health* 19, 2 (July 2022). [https://doi.org/10.5027/jmh-Vol19-Issue2\(2022\)art140](https://doi.org/10.5027/jmh-Vol19-Issue2(2022)art140)
- [26] Stanford University. 2021. Four causes for ‘Zoom fatigue’ and their solutions. <https://news.stanford.edu/2021/02/23/four-causes-zoom-fatigue-solutions/>
- [27] Faheem Zafari, Athanasios Gkelias, and Kin K Leung. 2019. A survey of indoor localization systems and technologies. *IEEE Communications Surveys & Tutorials* 21, 3 (2019), 2568–2599.